

ALVA, OK, USA

WMO: 722052

Lat: 36.773N

Lon: 98.670W

Elev: 449

StdP: 96.04

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.0	-17.7	0.8	-8.7	-15.0	1.1	-5.9	14.3	8.5	12.7	9.7	4.5	0	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	39.1	22.7	37.9	23.0	37.0	23.0	25.7	33.9	25.0	33.1	24.3	32.3	7.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.5	28.5	22.8	18.5	27.7	22.4	18.1	27.3	81.7	34.1	78.7	32.4	75.8	32.2	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.7	10.6	-16.4	41.8	3.2	1.6	-18.7	43.0	-20.6	44.0	-22.4	44.9	-24.7	46.1		
(5)			-16.9	26.4	3.0	1.1	-19.1	27.2	-20.9	27.9	-22.6	28.5	-24.8	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	2.7	4.2	9.9	14.0	20.3	26.6	28.6	27.1	23.4	16.1	8.8	3.3		
	DBStd	10.48	5.64	6.42	5.49	5.17	4.87	3.37	3.22	3.36	3.84	5.26	5.35	5.38		
	HDD10.0	811	232	178	70	20	1	0	0	0	0	12	84	213		
	HDD18.3	2192	485	396	266	145	36	1	1	0	5	105	288	465		
	CDD10.0	2808	6	16	66	142	321	497	578	529	401	200	47	7		
	CDD18.3	1147	0	0	3	16	98	247	319	271	156	35	2	0		
	CDH23.3	14004	2	9	42	231	1113	3146	4234	3141	1646	401	34	5		
	CDH26.7	7492	0	1	5	68	492	1763	2535	1706	779	136	4	2		
(14) Wind	WSAvg	5.0	4.9	5.1	5.5	5.6	4.8	5.4	4.6	4.5	4.9	5.1	5.0	4.7		
(15) Precipitation	PrecAvg	726	25	29	59	64	92	110	81	92	50	60	41	29		
	PrecMax	920	89	76	201	203	275	225	163	185	151	248	147	78		
	PrecMin	458	0	0	1	0	4	9	10	1	0	0	0	1		
	PrecStd	135	24	23	51	45	56	59	38	49	39	64	45	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	24.1	27.2	32.5	37.0	41.1	42.5	41.0	37.7	32.9	27.2	22.0		
		MCWB	11.8	12.5	15.7	18.7	20.5	21.5	23.1	23.7	22.2	20.3	15.9	14.2		
	2%	DB	17.7	21.1	24.1	27.9	33.9	38.0	40.0	38.7	35.2	30.2	23.7	17.4		
		MCWB	9.3	11.9	15.6	17.8	20.9	22.5	22.5	23.5	22.1	20.2	15.0	10.8		
	5%	DB	14.8	17.6	22.4	26.0	32.2	37.1	38.0	37.3	33.6	27.6	20.2	14.0		
		MCWB	7.9	10.7	14.9	17.1	20.6	22.8	23.0	23.7	22.2	18.2	13.5	8.6		
	10%	DB	12.3	14.0	19.9	23.7	29.0	35.0	37.1	34.8	31.4	25.1	17.6	12.3		
		MCWB	6.7	8.4	13.8	16.4	19.9	22.4	23.1	23.1	21.8	17.4	11.7	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	15.3	19.7	21.4	24.7	26.2	26.5	26.4	26.2	24.1	18.7	15.8		
		MCDB	19.4	20.0	24.1	26.9	31.5	34.7	36.0	34.2	33.4	28.2	22.7	20.0		
	2%	WB	10.3	13.3	18.0	19.9	23.4	25.1	25.5	25.5	24.4	22.3	16.5	12.2		
		MCDB	16.3	18.9	21.7	25.2	29.6	33.8	34.1	33.7	30.6	26.4	20.4	15.4		
	5%	WB	8.4	11.4	16.4	18.8	22.4	24.3	24.8	24.7	23.4	20.6	14.8	9.5		
		MCDB	14.5	17.6	19.8	24.1	28.2	32.7	33.1	33.0	29.5	25.8	18.8	13.7		
	10%	WB	6.8	9.2	14.5	17.4	21.3	23.6	24.1	24.0	22.6	18.9	12.9	7.6		
		MCDB	11.9	13.9	18.9	22.1	26.9	31.6	32.3	31.9	28.8	24.3	17.2	11.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	12.9	13.3	13.7	13.3	13.9	13.5	12.4	12.9	13.6	13.5	11.8		
		MCDBR	18.1	18.2	16.8	17.0	16.2	16.3	15.8	15.5	15.2	16.6	17.4	16.2		
	5% WB	MCWBR	11.1	10.9	9.8	9.4	6.4	4.8	4.3	4.4	5.3	8.2	10.2	10.6		
		MCWBR	17.1	16.8	13.3	14.7	13.4	14.2	13.4	12.8	12.4	13.6	14.7	14.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.282	0.291	0.333	0.365	0.386	0.412	0.414	0.413	0.376	0.341	0.299	0.278		
		taud	2.517	2.501	2.400	2.344	2.363	2.324	2.338	2.299	2.384	2.462	2.514	2.535		
	Ebn at Noon	Ebn at Noon	922	953	934	916	897	870	864	859	876	880	891	904		
		Edh at Noon	78	90	110	123	123	128	125	127	111	93	78	72		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.73	3.45	4.55	5.65	6.33	7.06	6.95	6.22	5.20	3.88	2.92	2.36		
	RadStd	RadStd	0.26	0.47	0.34	0.38	0.57	0.51	0.38	0.35	0.33	0.47	0.34	0.23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page